FitSense: A Machine Learning Approach to Predicting Product Fit

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***Abstract*—Online fashion retail faces persistently high return rates, many of which are caused by uncertainty in garment sizing and product fit. Because customers cannot physically try on items before purchase, they must rely on inconsistent size charts and subjective reviews, which often leads to dissatisfaction and expensive reverse logistics. This project presents FitSense, a supervised machine learning framework for predicting whether a clothing item will fit a customer as *small*, *fit*, or *large*. The study uses the UCSD Clothing Fit dataset, which contains user attributes such as height, weight, age, body type, and cup size, along with product-related information and self-reported fit feedback from the RentTheRunway platform. The proposed pipeline includes data cleaning, feature engineering, missing-value treatment, categorical encoding, class imbalance correction using SMOTENC, and comparative evaluation of multiple classification models. Logistic Regression, K-Nearest Neighbors, and Random Forest are evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Among the evaluated models, Random Forest provides the strongest overall performance and the best balance across the minority classes. The results show that features such as product size, body-related attributes, and derived BMI are informative predictors of fit outcome. This work demonstrates that machine learning can support more personalized size recommendation systems and help reduce fit-related returns in e-commerce.**

***Index Terms*—e-commerce, clothing fit prediction, machine learning, class imbalance, SMOTE, recommendation systems**

I. Introduction

The rapid growth of online fashion retail has created new opportunities for personalization and convenience, but it has also amplified the problem of product returns. A major cause of clothing returns is poor fit. Unlike in-store shopping, online customers cannot physically try on garments before purchase and must instead rely on size charts, product descriptions, and customer reviews. Because sizing standards vary across brands, categories, and platforms, the same nominal size may correspond to very different real-world fits. This mismatch contributes to customer dissatisfaction, increased operational costs, and inefficiencies in reverse logistics.

This project addresses the fit prediction problem as a three-class classification task. Given a customer's physical attributes and a product's descriptive information, the goal is to predict whether the garment will fit as small, fit, or large. Compared with a binary fit versus no-fit formulation, this representation is more practical because it also captures the direction of misfit, which is valuable for downstream recommendation and decision support. To study this problem, we use the UCSD Clothing Fit dataset introduced by Misra et al., which provides large-scale transaction-level data from ModCloth and RentTheRunway, including body measurements, product metadata, and user-reported fit outcomes.

The main contributions of this work are threefold. First, we develop a complete machine learning pipeline for fit prediction that includes preprocessing, feature engineering, imbalance handling, and model evaluation. Second, we compare several widely used classification methods, including Logistic Regression, K-Nearest Neighbors, and Random Forest, under a consistent evaluation framework. Third, we analyze which customer and product features are most informative for predicting fit outcomes and discuss how these insights can support more reliable and personalized e-commerce experiences.

II. Related Work

Product fit prediction lies at the intersection of recommender systems, apparel analytics, and applied machine learning. One of the most relevant prior studies is the work of Misra, Wan, and McAuley, who introduced the clothing fit dataset used in this project and proposed a latent factor framework for modeling fit semantics in a learned metric space. Their results showed that body-related attributes and product metadata contain meaningful signal beyond standard size labels, establishing fit prediction as an important and nontrivial recommendation problem.

Recent research has also explored richer body representations for garment fit modeling. Dik et al. proposed a framework that combines psychographic information and three-dimensional body measurements with artificial neural networks for fit prediction, and further used generative methods to visualize fitted garments. Their findings indicate that more detailed body descriptors can significantly improve predictive quality. Although the present project does not use 3D scan data, this line of work motivates the inclusion of derived body features such as BMI and categorical proxies such as body type and cup size.

Work in fashion recommendation has further shown that fit prediction is influenced by platform context and user behavior. Borgersen et al. studied recommendation in the

fashion rental setting and argued that content-based signals, including garment attributes and body measurements, are essential complements to collaborative filtering approaches. Their findings are particularly relevant to this project because the dataset combines two platforms with potentially different user expectations and fit patterns.

From a modeling perspective, prior studies in e-commerce classification have demonstrated the usefulness of both classical and ensemble learning methods. Shetty et al. compared multiple machine learning and deep learning approaches for e-commerce recommendation tasks and found that ensemble methods performed strongly under class imbalance. This supports the inclusion of Random Forest as a core ensemble method in the present study. In addition, Chawla et al. introduced SMOTE, which remains one of the most widely used methods for addressing class imbalance in supervised learning. Since the majority of clothing transactions are labeled as fit, while small and large are less frequent, SMOTE is a suitable choice for improving minority-class learning when applied carefully within the training folds only.

Another important distinction in the literature is the difference between structured tabular fit prediction and text-enhanced recommendation. While this project focuses on structured user and product metadata, review text may encode additional information about tightness, fabric stretch, garment cut, and user expectations. Prior work in sentiment-aware e-commerce prediction suggests that textual features can complement tabular signals, especially when structured attributes alone cannot fully explain user satisfaction. This motivates the inclusion of review-based representations as a future extension of the present system.

Taken together, the literature suggests four central themes that guide this project: body attributes are highly informative for fit prediction, ensemble models are strong baselines for tabular data, imbalance correction is necessary for reliable classification, and richer user-generated signals such as reviews may provide further gains in future work. Building on these findings, this paper develops a reproducible machine learning pipeline for multiclass clothing fit prediction using real-world e-commerce data.

III. Technical Approach

A. Dataset Description

This study uses the UCSD Clothing Fit dataset, which contains user feedback from two online fashion platforms: ModCloth and RentTheRunway. Each transaction record includes a fit label indicating whether the purchased item was perceived as small, fit, or large, along with user-specific and product-specific attributes. The most relevant fields used in this work include height, weight, age, body type, cup size, ordered clothing size, product category, rating, and anonymized user and item identifiers. The dataset provides a practical benchmark for fit prediction because it combines structured user characteristics with real-world self-reported fit outcomes at scale.

B. Data Preprocessing and Feature Engineering

The raw dataset requires substantial preprocessing before model training. Records with missing target labels are removed. For numerical attributes such as height, weight, and age, missing values are handled using median imputation, while categorical features such as body type and cup size are filled using mode imputation when appropriate. Rows with excessive missing information are excluded to maintain data quality. Height values originally represented as strings are converted into numeric inches, and BMI is computed from height and weight as a compact feature representing body composition. Clothing size is transformed into an ordinal representation, while categorical variables such as body type, category, and cup size are encoded for model compatibility. Continuous variables are standardized when required, particularly for distance-based models such as K-Nearest Neighbors.

A key challenge in this dataset is class imbalance. Most records are labeled as fit, while the small and large classes are underrepresented. To address this issue, the project applies Synthetic Minority Over-sampling Technique (SMOTE) during training. Importantly, oversampling is performed only within the training portion of each cross-validation split in order to avoid data leakage and preserve valid evaluation on unseen examples.

C. Data Cleaning and Representation

The original RentTheRunway records contain missing values and heterogeneous feature formats that must be standardized before training. After filtering invalid examples, the working dataset contains 158,219 records. Missing values in numerical fields were imputed using the median, while categorical fields such as body type and bust size were filled using the mode. Height values, originally stored in feet-and-inches string format, were converted into numeric inches. Weight values, originally represented in pounds, were converted to kilograms for consistency. Review dates were parsed into datetime format to ensure type correctness during preprocessing.

Several variables also required structural transformation. The original bust size field was decomposed into a numeric component and a categorical cup-size component. Clothing size was treated as an ordered feature, while body type, product category, and occasion-related variables were converted into one-hot encoded indicators. To reduce sparsity from rare categories, infrequent labels were grouped into an “uncommon” bucket before encoding. This representation allowed the pipeline to preserve structured user and product information while remaining compatible with standard tabular classifiers.

D. Derived Features and Class Rebalancing

In addition to the raw attributes, the pipeline introduces derived features intended to capture body composition more effectively. In particular, body mass index (BMI) was computed from user weight and height and included as a compact proxy for body shape. Since weight and height are individually

informative but also partially redundant, BMI provides a more interpretable summary feature for downstream models. Correlation analysis further showed strong positive relationships among weight, product size, and bust-size-related features, motivating the use of these variables as core predictors.

Class imbalance is a major challenge in this dataset because the majority of examples are labeled as *fit*, while *small* and *large* are much less frequent. To address this issue, Synthetic Minority Over-sampling Technique for Nominal and Continuous features (SMOTENC) was applied on the training split. After resampling, each class contained 107,733 examples, producing a balanced training distribution for KNN and Random Forest. This step substantially improved minority-class sensitivity compared with the baseline model trained directly on the imbalanced data.

E. Modeling Pipeline

The end-to-end pipeline begins with data ingestion from the RentTheRunway dataset, followed by cleaning, feature transformation, and exploratory analysis. After preprocessing, three classification algorithms are trained and compared under a consistent evaluation protocol: Logistic Regression, K-Nearest Neighbors, and Random Forest. Logistic Regression serves as an interpretable linear baseline trained on the original imbalanced data. KNN and Random Forest are both trained on the SMOTENC-resampled dataset to assess the impact of class balancing on minority-class prediction.

Hyperparameter tuning is performed using grid search with stratified 5-fold cross-validation. For KNN, the search space covers neighborhood size (k from 2 to 19) and weighting strategy (uniform or distance-weighted). For Random Forest, the grid includes the number of estimators, maximum tree depth, feature selection strategy, and minimum samples per split and leaf. This procedure ensures that model comparisons reflect optimized configurations rather than default parameter settings.

F. Evaluation Strategy

All models are evaluated using stratified 5-fold cross-validation to preserve the class distribution across splits. Performance is measured using accuracy, precision, recall, and F1-score. Because the dataset is imbalanced, special attention is given to macro-averaged metrics and to classification quality on the minority classes, especially the small and large labels. In addition to aggregate scores, the final analysis includes confusion matrices and feature importance interpretation for the strongest-performing models.

IV. Experimental Results

A. Experimental Setup

All experiments use the RentTheRunway subset of the UCSD Clothing Fit dataset, which contains 158,219 records after preprocessing. The feature set includes user weight (kg), height (inches), product size, age, bust size (numeric and category), BMI, and one-hot encoded body type, product category, and occasion fields. SMOTENC is applied to the training split to balance the three fit classes (*small*, *fit*, *large*), yielding 107,733 samples per class in the resampled training set. An 80/20 train-test split is used throughout.

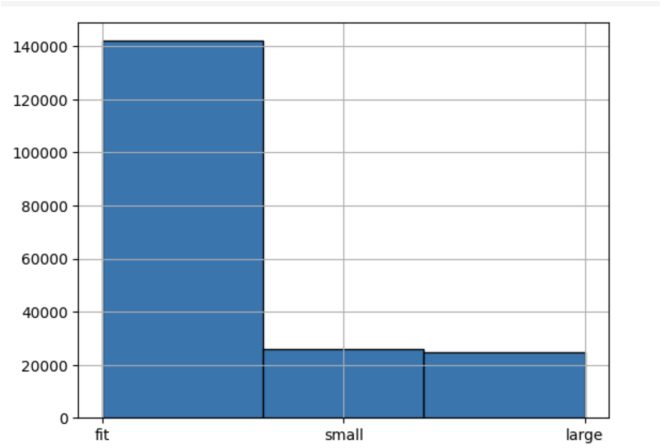

Fig. 1. Distribution of fit labels showing class imbalance

B. Class Distribution and EDA Findings

The distribution of the target variable reveals a clear class imbalance in the dataset, as shown in Fig. 1. The majority of samples are labeled as *fit*, while the *small* and *large* classes are significantly underrepresented. Approximately 68% of the records belong to the *fit* class, with the remaining portion split between *small* and *large*. This imbalance is important because models trained directly on such data tend to favor the majority class, often at the expense of correctly identifying minority outcomes.

To address this issue, SMOTENC is applied to balance the class distribution in the training set. By generating synthetic examples for the minority classes, the model is encouraged to learn more representative decision boundaries. Without such correction, baseline models such as Logistic Regression achieve reasonable overall accuracy but fail to capture minority-class behavior effectively.

In addition to class imbalance, exploratory analysis of numerical features provides insight into the structure of the prediction problem. Fig. 2 shows the correlation matrix for the numerical attributes used in this study. A strong positive correlation is observed between user weight and product size, as well as between product size and bust-size-related features. These relationships indicate that body composition and garment sizing are closely linked and jointly influence perceived fit.

Further analysis of feature distributions grouped by fit label reveals consistent trends. Higher product sizes and body measurements are more frequently associated with *large* outcomes, while lower values tend to correspond to *small* outcomes. These observations confirm that the fit prediction task is driven by structured relationships between user characteristics and garment sizing rather than random variation.

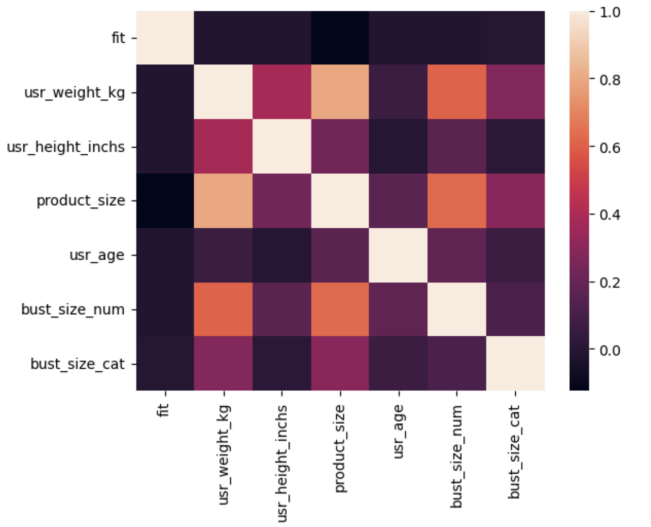

Fig. 2. Correlation matrix of numerical features

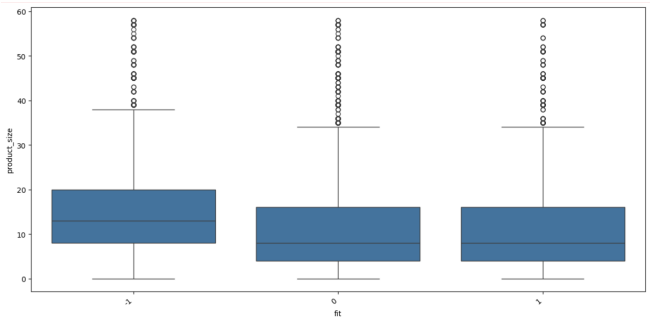

Fig. 3. Distribution of product size across fit categories

C. Fit Trends by Product Size

Product size was one of the most important structured features in the dataset. As shown in Fig. 3, the distribution of product size varies across the three fit outcomes. Items labeled as *large* tend to appear at higher product sizes, while *small* outcomes are more concentrated around lower product sizes. The *fit* class lies between these two patterns, suggesting that ordered size information provides meaningful signal for distinguishing between correct and incorrect fit outcomes.

This trend supports the use of product size as a core predictor in the modeling pipeline. It also shows that fit prediction is not only a customer-level problem, but also depends on the interaction between customer body attributes and garment sizing.

D. Model Performance

Table I summarizes the performance of the three evaluated classifiers. Logistic Regression was trained on the original imbalanced data as an interpretable baseline prior to oversampling. K-Nearest Neighbors and Random Forest were both trained on the SMOTE-resampled dataset.

TABLE I
CLASSIFICATION PERFORMANCE ON THE TEST SET

Model	Acc.	Macro P	Macro R	Macro F1
Logistic Regression	0.69	0.52	0.37	0.35
KNN (k=2, SMOTE)	0.69	0.70	0.69	0.68
Random Forest (SMOTE)	0.76	0.76	0.76	0.75

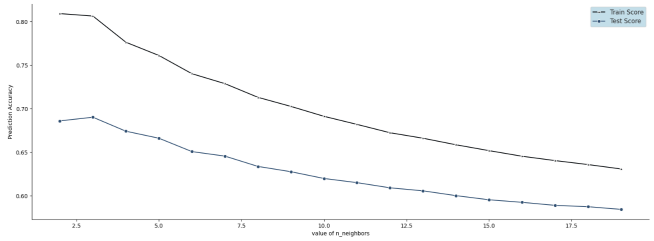

Fig. 4. Training and testing accuracy of KNN for different values of k

Logistic Regression achieves an overall accuracy of 69% but produces a macro F1-score of only 0.35, driven by near-complete failure on the minority classes: recall of 0.11 for *small* and 0.02 for *large*. This result confirms that a linear model trained on imbalanced data is insufficient for this task and serves as a lower-bound reference.

KNN with k=2 after SMOTE resampling achieves 69% accuracy and a substantially improved macro F1-score of 0.68. Per-class F1-scores are 0.75 for *small*, 0.59 for *fit*, and 0.70 for *large*, indicating that oversampling meaningfully recovers minority-class sensitivity. Hyperparameter tuning via 5-fold GridSearchCV identified k=6 with distance-weighted voting as the optimal configuration.

To evaluate the sensitivity of KNN to the neighborhood size, the model was tested across multiple values of k. Fig. 4 shows that smaller neighborhood sizes achieved stronger test performance, with $k = 2$ producing the best result before further hyperparameter tuning. As k increased, the model became smoother but less responsive to local patterns in the balanced dataset.

This behavior is consistent with the bias-variance tradeoff in KNN. Smaller values of k capture local decision boundaries more aggressively, while larger values reduce variance but may underfit class-specific patterns. In this project, KNN served as a useful non-parametric baseline, but it did not outperform Random Forest.

Random Forest with 50 estimators and SMOTE resampling achieves the best overall performance, reaching 76% accuracy and a macro F1-score of 0.75. Per-class F1-scores are 0.79 for *small*, 0.68 for *fit*, and 0.79 for *large*. This balanced performance across all three classes, including the minority categories, makes Random Forest the strongest model in this evaluation.

E. Error Analysis and Feature Importance

Fig. 5 presents the confusion matrix for the Random Forest classifier, which achieved the strongest overall performance among the evaluated models. Compared with the Logistic

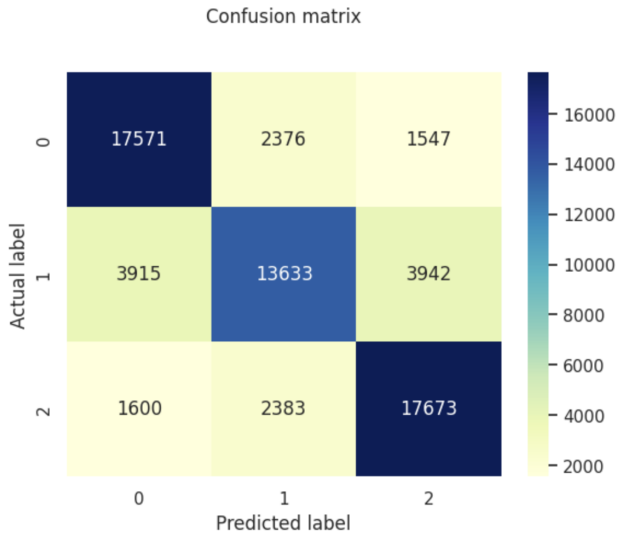

Fig. 5. Confusion matrix for the Random Forest classifier

Regression baseline, Random Forest preserves much better recognition of the minority *small* and *large* classes. This indicates that the combination of SMOTENC-based balancing and nonlinear ensemble learning is effective for this prediction task.

Most remaining errors occur near the boundary between the *fit* class and the two misfit classes. This is expected because fit perception is subjective and can depend on factors not fully captured by the structured features, such as garment fabric, stretch, cut, and individual user preference. Even with these limitations, the confusion matrix shows that Random Forest provides a relatively balanced classification pattern across all three labels.

V. Conclusion

This paper presented FitSense, a supervised machine learning pipeline for predicting clothing fit outcomes in online retail as a three-class classification problem. Using the UCSD Clothing Fit dataset from the RentTheRunway platform, the study developed a preprocessing pipeline that includes missing-value imputation, feature engineering, BMI derivation, categorical encoding, and SMOTENC-based class imbalance correction. Three classification models were evaluated: Logistic Regression as a baseline, K-Nearest Neighbors, and Random Forest.

The results demonstrate that class imbalance correction is a critical prerequisite for meaningful minority-class prediction in this domain. Logistic Regression without oversampling achieved a macro F1-score of only 0.35 despite 69% overall accuracy, while KNN and Random Forest trained on SMOTE-resampled data reached macro F1-scores of 0.68 and 0.75 respectively. Random Forest emerged as the strongest model, achieving 76% accuracy with balanced per-class performance, and its feature importance rankings confirm that product size, BMI, and bust-related attributes are the primary drivers of fit prediction.

Beyond model comparison, the results also highlight the practical value of interpretable feature trends. The strongest-performing models consistently relied on product size, BMI, and bust-related features, indicating that meaningful fit signals can be extracted from relatively simple structured metadata. This is encouraging for real-world deployment because such features are easier to collect and standardize than expensive alternatives such as 3D body scans or full virtual try-on systems.

This work has several limitations. The current evaluation is conducted on the RentTheRunway subset only; incorporating the ModCloth data in a joint or stratified analysis is a natural next step. The pipeline also does not yet exploit the free-text review field, which may contain latent signals about fit perception. Future work could integrate TF-IDF or embedding-based features from review text, evaluate gradient boosting methods such as XGBoost and LightGBM, and explore category-specific submodels to address the differing fit dynamics observed across product types.

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Project Information

Code Availability: The source code and notebooks used in this project are available at `https://github.com/vandanshah17/CMPE_255_FitSense`.

Author Contributions: Vandan Sanket Shah led data cleaning and preprocessing. Neeraja Abhinav Buch conducted exploratory data analysis. Mokshit Chopra developed and implemented the classification models. Tanisha Ashishbhai Dave performed testing and evaluation. All authors contributed to project design, interpretation of results, and preparation of the final manuscript. All authors are enrolled in CMPE 255, Section 01, San José State University.

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